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Major Project

1010073492

LLM Security Testing Platform

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Introduction

- Large Language Models (LLMs) like ChatGPT, Claude, and Gemini are being rapidly integrated into enterprise applications - chatbots, customer service, code generation, and decision-making systems.
- This integration introduces a new class of security vulnerabilities unique to AI systems, particularly Prompt Injection attacks.
- Prompt Injection allows attackers to manipulate LLM behavior by crafting malicious inputs that override system instructions, bypass safety guardrails, or extract sensitive information.
- PromptShield is a security testing platform that enables developers and security teams to test their LLM applications against known attack vectors and implement effective defenses.
- Research Domain: AI Security, Cybersecurity, Application Security

Background and Motivation

Background:

- Traditional security testing tools (OWASP ZAP, Burp Suite) were designed for web applications and are inadequate for LLM systems.
- The OWASP Top 10 for LLM Applications (2023) identified Prompt Injection as the #1 security risk for LLM-integrated applications [1].
- 86% of 36 real-world LLM applications tested were found vulnerable to prompt injection attacks (HouYi Research, 2023) [2].

Motivation:

- No comprehensive, open-source tool exists for testing LLM security.
- Organizations deploying AI lack visibility into their vulnerability exposure against prompt-based attacks.
- Personal interest in bridging cybersecurity expertise with emerging AI security challenges.

Relevance and Importance

Why This Research Matters:

1. Industry Need:

- Market for AI security tools projected to reach \$60B by 2028 [3]
- Every company deploying LLMs needs security testing capabilities

2. Academic Contribution:

- Novel approach combining multiple detection techniques
- Practical open-source contribution to the security community

3. Who Benefits:

- Developers building LLM-powered applications
- Security teams conducting AI red-teaming or ensuring AI safety.

4. Research Gap:

- Existing research focuses on attacks, not detection tools
- No standardized testing framework for LLM vulnerabilities

Literature Review

Key Research Papers:

[1] OWASP Top 10 for LLM Applications (2023)

- Established first standardized vulnerability taxonomy for LLMs
- Identified 10 critical risks including prompt injection, data poisoning, and sensitive information disclosure

[2] "Prompt Injection Attack Against LLM-Integrated Applications"

- Liu et al. (2023) - arXiv:2306.05499. Introduced HouYi attack framework
- Tested 36 applications, found 31 vulnerable (86%). Validated by 10 vendors including Notion

[3] "Ignore This Title and HackAPrompt" - Schulhoff et al. (2023)

- Created comprehensive taxonomy of prompt injection attacks
- Categorized attacks into direct and indirect injection

Current Status: Active research area with weekly new publications

Objectives

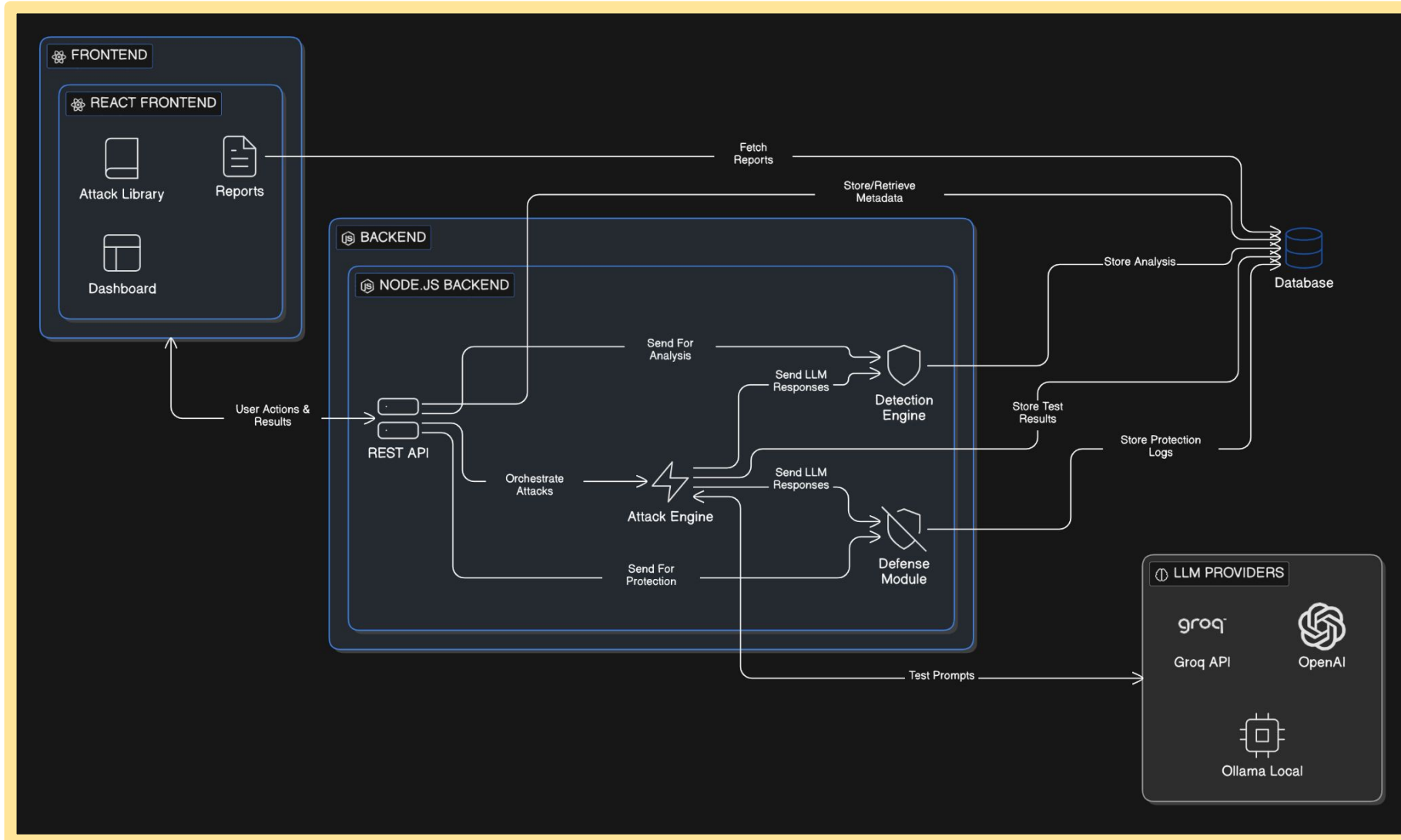
Primary Objectives:

1. Develop a web-based platform for automated LLM security testing
2. Create a comprehensive attack library covering:
 - Prompt Injection (direct and indirect)
 - Jailbreak attempts
 - Data/System Prompt Extraction
 - Context Manipulation
3. Implement automated vulnerability detection algorithms that analyze LLM responses for security violations
4. Design a defense testing module to evaluate protection strategies:
 - Input sanitization
 - Prompt hardening
 - Output filtering
5. Generate actionable security reports with risk scores and remediation recommendations
6. Open-source the platform for community contribution and research

Required Tools and Technology

Category	Component/Tool	Details
Frontend	Build Tool	React.js with Vite
	Components & Styling	Shadcn/UI & Tailwind CSS
	Visualization	Chart.js/Recharts
Backend	Framework	Node.js with Express.js
	Architecture	REST API
LLM Integration	Main & Optional API	Groq API & OpenAI API
	Local Testing	Ollama
Deployment	Hosting (Frontend/Backend)	Vercel & Render/Railway
	Version Control & CI/CD	Git & GitHub
Development Tools	Tools, IDEs	VS Code, Antigravity, Postman

Required Tools and Technology



Project Approach

Phase 1: Research & Design (Week 1)

- Literature review of LLM vulnerabilities
- Study OWASP Top 10 for LLM Applications
- Design system architecture and API structure

Phase 2: Core Development (Week 2-3)

- Backend API development (Node.js)
- Attack implementation (50+ attack vectors)
- LLM integration layer for multiple providers

Phase 3: Detection Engine (Week 3-4)

- Pattern-based detection algorithms
- Semantic analysis of responses
- Confidence scoring system

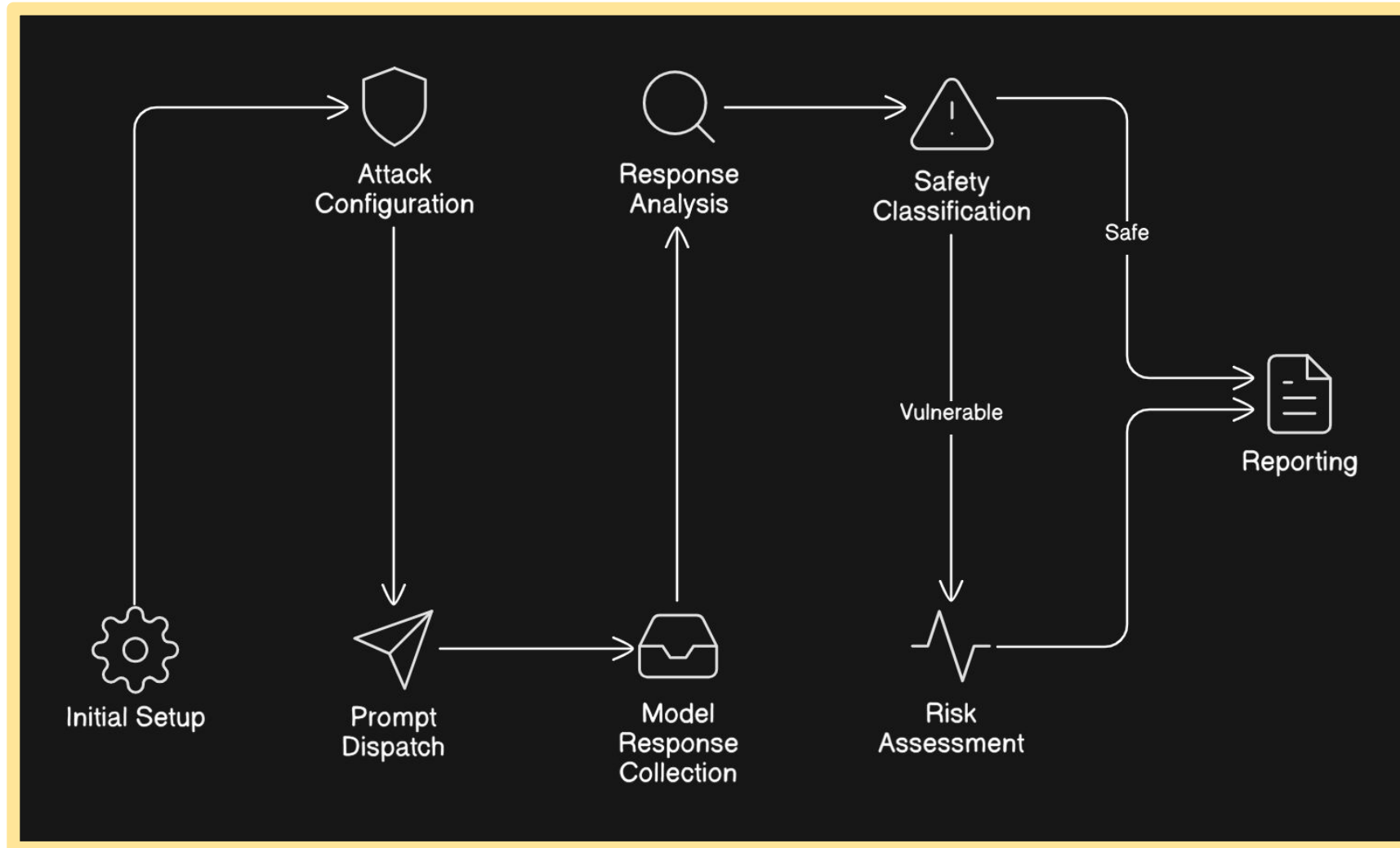
Phase 4: Frontend & Integration (Week 4-5)

- Dashboard development (React)
- Real-time testing interface
- Report generation module

Phase 5: Testing & Documentation (Week 5-6)

- End-to-end testing
- Security validation
- Research paper writing

Project Working Method



Bibliography

- [1] OWASP Foundation. ["OWASP Top 10 for Large Language Model Applications v1.1."](#) OWASP, 2023.
- [2] Liu, Y., Deng, G., Li, Y., et al. ["Prompt Injection attack against LLM-integrated Applications."](#) arXiv preprint arXiv:2306.05499, 2023.
- [3] Schulhoff, S., Pinto, J., et al. ["Ignore This Title and HackAPrompt: Exposing Systemic Vulnerabilities of LLMs through a Global Scale Prompt Hacking Competition."](#) arXiv:2311.16119, 2023.
- [4] Wei, A., Haghtalab, N., Steinhardt, J. ["Jailbroken: How Does LLM Safety Training Fail?"](#) arXiv:2307.02483, 2023.
- [5] Greshake, K., et al. ["Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection."](#) arXiv:2302.12173, 2023.